Category Theory for Mathematical Physics — Exercise Sheet 1

Categories: a theory of systems and processes

Variant: Hints and answers

Conventions as in Lecture 1: in $g \circ f$ the morphism f acts first; BG has one object $*$ with $\mathrm{Hom}(*,*) = G$ and $g \circ h = gh$; $\mathrm{Rep}(G)$ has the finite-dimensional complex representations and all intertwiners. A *homomorphism* of groups preserves products (and hence the identity). The cyclic group $\mathbb{Z}_n$ is written additively. In Exercise 3, $\omega = e^{2\pi i/3}$.

Exercise 1 (Categories, non-examples, and groups). (i) Take as objects the natural numbers $0, 1, 2, \ldots$, with exactly one morphism $m \to n$ when $m \leq n$ and none otherwise. Show that this is a category, and find its isomorphisms.

(ii) Take as objects the finite-dimensional complex vector spaces, and as morphisms $V \to W$ the linear maps of rank at most 1, composed as maps. Show that a composite of two such maps is again one. Show that this is not a category, and find the objects V that have an identity morphism.

(iii) Show that a functor $BG \to BH$ is exactly a homomorphism $G \to H$. How many functors are there $B\mathbb{Z}_4 \to B\mathbb{Z}_2$, and how many $B\mathbb{Z}_2 \to B\mathbb{Z}_3$?

(iv) Let $\varphi, \psi \colon G \to H$ be homomorphisms, read as functors $BG \to BH$. Show that a natural transformation $\varphi \Rightarrow \psi$ is an element $h \in H$ with $\psi(g) = h\,\varphi(g)\,h^{-1}$ for all $g \in G$, and that it is always a natural isomorphism. There are 10 homomorphisms $S_3 \to S_3$; how many classes do they form up to natural isomorphism?

Hint. In (i) and (iii) each axiom becomes a familiar property: of $\leq$ in (i), of a homomorphism in (iii). In (ii), an identity $e \colon V \to V$ must satisfy $e \circ f = f$ for every morphism f into V; try f of rank one with image any chosen line. In (iv), a natural transformation between functors from BG has a single component; write its naturality square for one morphism g, and sort the homomorphisms $S_3 \to S_3$ by their kernels.

Answer. (i) A category; its only isomorphisms are the identities. (ii) Composites have rank at most 1, but an object V has an identity exactly when $\dim V \leq 1$, so it is not a category. (iii) 2 functors $B\mathbb{Z}_4 \to B\mathbb{Z}_2$ and 1 functor $B\mathbb{Z}_2 \to B\mathbb{Z}_3$. (iv) 3 classes: the trivial homomorphism, the three onto a subgroup of order 2, and the six automorphisms.

Exercise 2 (G-sets are functors $BG \to \mathrm{Set}$). A *G-set* is a set X with an action $g \cdot x$, such that $1 \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$; a map $f \colon X \to Y$ of G-sets is *equivariant* if $f(g \cdot x) = g \cdot f(x)$.

(i) Show that a functor $BG \to \mathrm{Set}$ is a G-set, and that a natural transformation between two such functors is an equivariant map.

(ii) Let $G = S_3$, let $X = \{1, 2, 3\}$ with the permutation action, and let $Y = \{+, -\}$ on which the even permutations act trivially and the odd ones swap $+$ and $-$. Count the equivariant maps $X \to X$, $X \to Y$, $Y \to X$ and $Y \to Y$.

(iii) For a finite G-set X, let $\mathbb{C}[X]$ have basis e_x ($x \in X$) and $\rho(g)e_x = e_{g \cdot x}$. Show that $\mathbb{C}[X]$ is a representation and that an equivariant $f \colon X \to Y$ gives an intertwiner $\mathbb{C}[f] \colon e_x \mapsto e_{f(x)}$, and that $X \mapsto \mathbb{C}[X]$, $f \mapsto \mathbb{C}[f]$ is a functor from finite G-sets to $\mathrm{Rep}(G)$.

Hint. (i) is the proof of Theorem 1.9 with Set in place of $\mathrm{Vect}_{\mathrm{fd}}$. In (ii), if X is a single orbit, an equivariant map is determined by the image of one point x_0; which images are allowed? Look at the elements of G that fix x_0.

Answer. (i) A functor is a set with an action of G, and naturality is equivariance. (ii) 1, 0, 0 and 2. (iii) Yes: $\mathbb{C}[X]$ is a representation, $\mathbb{C}[f]$ an intertwiner, and $\mathbb{C}[f' \circ f] = \mathbb{C}[f'] \circ \mathbb{C}[f]$, $\mathbb{C}[\mathrm{id}] = \mathrm{id}$.

Exercise 3 (Restriction is a functor, faithful and not full). Let $H \leq G$ be a subgroup and $\iota\colon H \to G$ the inclusion.

(i) Show that restriction, $\mathrm{Res}(V,\rho) = (V, \rho \circ \iota)$ and $\mathrm{Res}(T) = T$, is a functor $\mathrm{Rep}(G) \to \mathrm{Rep}(H)$, and that on objects, read through Theorem 1.9, it is composition with the functor $B\iota\colon BH \to BG$.

(ii) Show that Res is faithful.

(iii) Let $G = S_3$, $H = A_3$, and $V = \{x \in \mathbb{C}^3 : x_1+x_2+x_3 = 0\}$ with the permutation action, the standard representation. Show that $\dim \mathrm{Hom}_{S_3}(V,V) = 1$ and $\dim \mathrm{Hom}_{A_3}(\mathrm{Res}\, V, \mathrm{Res}\, V) = 2$. Conclude that Res is not full.

(iv) Write down an intertwiner of $\mathrm{Res}\, V$ that does not come from $\mathrm{Hom}_{S_3}(V,V)$.

Hint. (i) and (ii) take one line each. In (iii), V is irreducible for S_3; for A_3, find the eigenvalues of a 3-cycle c on V from the trace of the permutation matrix, and ask which matrices commute with $\rho(c)$.

Answer. (i) An intertwiner for G is one for H; a representation read as a functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ is restricted by composing it with $B\iota$. (ii) Res is the identity on morphisms. (iii) 1 and 2; not full. (iv) The projection $P_\omega = \frac{1}{3}\left(\mathrm{id} + \bar{\omega}\rho(c) + \omega\rho(c)^2\right)$ onto the ω-eigenline of $c = (1\,2\,3)$, $\omega = e^{2\pi i/3}$.

Exercise 4 (The dual over the reals). Let $\mathrm{Vect}_{\mathbb{R}}^{\cong}$ be the groupoid of finite-dimensional real vector spaces and linear isomorphisms, with $D(V) = V^*$ and $D(f) = (f^*)^{-1}$ as in Lecture 1.

(i) Show that no family of linear isomorphisms $\theta_V : V \to V^*$ is a natural transformation $\mathrm{id} \Rightarrow D$ on $\mathrm{Vect}_{\mathbb{R}}^{\cong}$.

(ii) On the groupoid of finite-dimensional real inner-product spaces and orthogonal maps, show that $R_V(u) = \langle u, \cdot \rangle$ is a natural isomorphism $\mathrm{id} \Rightarrow D$. Why does this contradict neither (i) nor Theorem 1.11?

(iii) For which real numbers λ does the test $f = \lambda\, \mathrm{id}_V$ prove (i)? What plays the role of λ in the lecture's proof for unitary maps, and why is there no such scalar among the orthogonal maps?

Hint. Naturality reads $f^* \circ \theta_W \circ f = \theta_V$, as in the lecture. Compute both sides for $f = \lambda\, \mathrm{id}_V$, and for (ii) evaluate $f^*(R_W(fu))$ on a vector v.

Answer. (i) With $f = 2\, \mathrm{id}_V$, naturality gives $4\theta_V = \theta_V$, so $\theta_V = 0$. (ii) $f^*(R_W(fu))(v) = \langle fu, fv \rangle = \langle u, v \rangle = R_V(u)(v)$; the real inner product is bilinear, so R_V is linear, and 2 id is not orthogonal. (iii) Exactly those with $\lambda^2 \neq 1$ (and $\lambda \neq 0$); the lecture uses $\lambda = i$; the only orthogonal scalars are ± 1.

Exercise 5 (Skeletons). (i) Let FinBij be the groupoid of finite sets and bijections, and S the category with objects $0, 1, 2, \ldots$, $\mathrm{Hom}(n,n) = S_n$ and no morphisms $m \to n$ for $m \neq n$. Show that S is a skeleton of FinBij, and that the two are not isomorphic.

(ii) Let $\mathcal{G}$ be a groupoid in which any two objects are isomorphic, and x an object. Show that $\mathcal{G} \simeq B\,\mathrm{Aut}(x)$, where $\mathrm{Aut}(x) = \mathrm{Hom}(x,x)$. Apply this to the groupoid of the two-element subsets of $\{1,2,3\}$ with bijections.

(iii) In the proof of Theorem 1.16 the bases were chosen with $b_{\mathbb{C}^n} = \mathrm{id}$. Keep every choice but one: $b_{\mathbb{C}^2} = \mathrm{diag}(1,2)$. Compute $G(M(A))$ for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Is $G \circ M = \mathrm{id}$? Are M and G still an equivalence?

(iv) Lecture 1's functor "forgetting crossings" goes from the braid groupoid Br to the category S of (i). Show that it is full and not faithful. You may use one fact about braids: two braids on the same strands are equal in Br only if they have the same *exponent sum*, the number of positive crossings minus the number of negative ones, where a crossing is positive when the strand going up to the right passes over.

Hint. For (i) and (ii), use Proposition 1.17: check full, faithful and essentially surjective, then argue as in Example 1.18. In (iii), $G(f) = b_W^{-1} \circ f \circ b_V$; look for a natural isomorphism between $G \circ M$ and the identity. In (iv), every permutation is a product of transpositions of neighbours; for faithfulness, compare one crossing taken twice with no crossing.

Answer. (i) $n \mapsto \{1, \dots, n\}$ is full, faithful and essentially surjective, and S is skeletal; not isomorphic, since FinBij has distinct isomorphic objects. (ii) The functor $B\operatorname{Aut}(x) \to \mathcal{G}$, $* \mapsto x$, is an equivalence; the example is equivalent to BS_2 (three objects, each Hom set of size 2). (iii) $G(M(A)) = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \neq A$, so $G \circ M \neq \mathrm{id}$; still an equivalence, with $G \circ M \cong \mathrm{id}$ through the components $b_{\mathbb{C}^n}^{-1}$. (iv) Full: each $(i\ i{+}1)$ lifts to one crossing. Not faithful: a crossing taken twice and the identity braid on two strands both forget to the identity permutation, but their exponent sums are ± 2 and 0.

Exercise 6 ⋆. *Natural transformations of the identity of* $\mathrm{Rep}(G)$. Let G be finite, $\mathbb{C}[G]$ its group algebra (basis $g \in G$, product extended linearly), and ρ_V extended linearly to $\mathbb{C}[G]$. A natural transformation $\xi\colon \mathrm{id}_{\mathrm{Rep}(G)} \Rightarrow \mathrm{id}_{\mathrm{Rep}(G)}$ is an intertwiner $\xi_V\colon V \to V$ for each representation V, with $T \circ \xi_V = \xi_W \circ T$ for every intertwiner $T\colon V \to W$.

(i) For z in the centre $Z(\mathbb{C}[G])$, show that $\xi_V = \rho_V(z)$ is such a natural transformation.

(ii) Conversely, let ξ be one, let $R = \mathbb{C}[G]$ be the regular representation, $\rho_R(g)a = ga$, and put $z = \xi_R(1)$. Draw the naturality of ξ for an intertwiner $T\colon R \to V$ as a box sliding through ξ, in the style of Lecture 1's Equation (1.2). Show that z is central and that $\xi_V = \rho_V(z)$ for every V.

(iii) Show that $z = \sum_g z_g g$ is central exactly when z_g is constant on conjugacy classes. Natural transformations of $\mathrm{id}_{\mathrm{Rep}(G)}$ are added and scaled component by component, and composed as in Lemma 1.8. Conclude that they form an algebra isomorphic to $Z(\mathbb{C}[G])$, of dimension the number of conjugacy classes, and give that number for S_3 and for the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Hint. In (ii), for $v \in V$ the map $T_v\colon R \to V$, $a \mapsto \rho_V(a)v$, is an intertwiner; write naturality for T_v at $a = 1$. For centrality, compute $\xi_R(g)$ in two ways: from ξ_R being an intertwiner, and from (ii)'s formula with $V = R$.

Answer. (i) $\rho_V(z)$ commutes with every $\rho_V(g)$, and every intertwiner commutes with it. (ii) $\xi_V(v) = \rho_V(z)v$, and $gz = \xi_R(g) = zg$. (iii) $\dim Z(\mathbb{C}[G])$ is the number of conjugacy classes: 3 for S_3, 5 for Q_8.